Model Architecture: Qwen3-0.6B

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1 Model Overview

Model name	Qwen3-0.6B
Transformers class	`Qwen3ForCausalLM`
Path	`/models/Qwen3-0.6B`
Total parameters	596,049,920 (596.05M)
Weight dtype	bfloat16
Hidden size d	1,024
Decoder layers L	28
Query heads H	16
Key/value heads H_{kv}	8
Head dim d_h	128
MLP width d_{ff}	3,072
Vocabulary V	151,936
Max context length	40,960
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	SiLU
RoPE base θ	1,000,000

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (1024)
V	vocabulary size (151,936)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 28 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$
\begin{aligned}
h_0 &= \text{Embed}(x) \\
h_L &= \left(\mathcal{B}_{\text{F}}^{\circ 28}\right)(h_0) \\
y &= W_E^{\top}\ \text{RMSNorm}_{\text{final}}(h_L)
\end{aligned}
$$

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\text{F}}$	Full (global) attention block	28	15,730,944	0–27

Language-model decoder (28 layers). layers 0–27: $[\mathcal{B}_{\text{F}}]$ repeated 28×.

2.2 Block functions

$\mathcal{B}_{\mathrm{F}}$ — Full (global) attention block.

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u && H = 16 \text{ heads of size } d_h = 128\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1; \ldots; O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\mathrm{SiLU}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\big) && \text{gated MLP, width } 3072
\end{aligned}
$$

3 Forward Pass in Pseudocode

Algorithm 1 Qwen3-0.6B: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow \mathrm{Embed}(x)$ ▷ look up one vector per token: $B{\times}T{\times}1024$
for $r \leftarrow 0$ **to** 27 **do** ▷ layers 0–27: pattern of 1 repeated 28 times
  $h \leftarrow \mathcal{B}_{\mathrm{F}}(h)$ ▷ Full (global) attention block
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$
$y \leftarrow h\, W_E^{\top}$ ▷ logits $B{\times}T{\times}151,936$, embedding matrix re-used
return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 1024$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^{\top}$ ▷ $B{\times}T{\times}16{\times}128$ (16 query heads)
$K \leftarrow uW_K^{\top};\ V \leftarrow uW_V^{\top}$ ▷ $B{\times}T{\times}8{\times}128$ (8 key/value heads, each shared by 2 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\, QK^{\top} + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}128$
$m \leftarrow O\, W_O^{\top}$ ▷ concatenate heads, project back to $B{\times}T{\times}1024$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^{\top}) \odot uW_{\mathrm{up}}^{\top}\big)W_{\mathrm{down}}^{\top}$ ▷ expand to 3072, gate, shrink back to 1024
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	151936×1024	155,582,464
`model.norm.weight`	Qwen3RMSNorm	1024	1,024
`lm_head.weight` (tied to `model.embed_tokens.weight`)	Linear	151936×1024	0 (shared)

4.2 Language-model decoder: `model.layers`

Table 3: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 0–27; weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×1024	2,097,152
`self_attn.k_proj.weight`	Linear	1024×1024	1,048,576
`self_attn.v_proj.weight`	Linear	1024×1024	1,048,576
`self_attn.o_proj.weight`	Linear	1024×2048	2,097,152
`self_attn.q_norm.weight`	Qwen3RMSNorm	128	128
`self_attn.k_norm.weight`	Qwen3RMSNorm	128	128
`mlp.gate_proj.weight`	Linear	3072×1024	3,145,728
`mlp.up_proj.weight`	Linear	3072×1024	3,145,728
`mlp.down_proj.weight`	Linear	1024×3072	3,145,728
`input_layernorm.weight`	Qwen3RMSNorm	1024	1,024
`post_attention_layernorm.weight`	Qwen3RMSNorm	1024	1,024
Per block			**15,730,944**
× 28 blocks			**440,466,432**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
LANGUAGE MODEL	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).

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Layer (appears as)	What it does
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 5: Parameters by component

Component	Parameters	% of total
MLP / MoE	264,241,152	44.3%
Attention / token mixer	176,167,936	29.6%
Token embedding	155,582,464	26.1%
Norms	58,368	0.0%
Total	**596,049,920**	**100.0%**

Table 6: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	28	15,730,944	440,466,432
Modules outside stacks	1	–	155,583,488
Total			**596,049,920**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	151936 × 1024	bfloat16	155,582,464
`model`	Qwen3Model	–	–	–
`model.embed_tokens.weight`	Embedding	151936 × 1024	bfloat16	155,582,464
`model.layers.0`	Qwen3DecoderLayer	–	–	–
`model.layers.0.input_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.0.mlp`	Qwen3MLP	–	–	–
`model.layers.0.mlp.down_proj.weight`	Linear	1024 × 3072	bfloat16	3,145,728
`model.layers.0.mlp.gate_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.0.mlp.up_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.0.post_attention_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.0.self_attn`	Qwen3Attention	–	–	–
`model.layers.0.self_attn.k_norm.weight`	Qwen3RMSNorm	128	bfloat16	128
`model.layers.0.self_attn.k_proj.weight`	Linear	1024 × 1024	bfloat16	1,048,576
`model.layers.0.self_attn.o_proj.weight`	Linear	1024 × 2048	bfloat16	2,097,152
`model.layers.0.self_attn.q_norm.weight`	Qwen3RMSNorm	128	bfloat16	128
`model.layers.0.self_attn.q_proj.weight`	Linear	2048 × 1024	bfloat16	2,097,152
`model.layers.0.self_attn.v_proj.weight`	Linear	1024 × 1024	bfloat16	1,048,576
`model.layers.1`	Qwen3DecoderLayer	–	–	–
`model.layers.1.input_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.1.mlp`	Qwen3MLP	–	–	–
`model.layers.1.mlp.down_proj.weight`	Linear	1024 × 3072	bfloat16	3,145,728
`model.layers.1.mlp.gate_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.1.mlp.up_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.1.post_attention_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.1.self_attn`	Qwen3Attention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.1.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.1.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.1.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.1.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.1.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.1.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.2	Qwen3DecoderLayer	–	–	–
model.layers.2.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.2.mlp	Qwen3MLP	–	–	–
model.layers.2.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.2.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.2.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.2.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.2.self_attn	Qwen3Attention	–	–	–
model.layers.2.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.2.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.2.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.2.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.2.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.2.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.3	Qwen3DecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.3.mlp	Qwen3MLP	–	–	–
model.layers.3.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.3.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.3.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.3.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.3.self_attn	Qwen3Attention	–	–	–
model.layers.3.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.3.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.3.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.3.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.3.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.3.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.4	Qwen3DecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.4.mlp	Qwen3MLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.4.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.4.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.4.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.4.self_attn	Qwen3Attention	–	–	–
model.layers.4.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.4.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.4.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.4.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.4.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.4.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.5	Qwen3DecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.5.mlp	Qwen3MLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.5.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.5.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.5.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.5.self_attn	Qwen3Attention	–	–	–
model.layers.5.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.5.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.5.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.5.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.5.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.5.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.6	Qwen3DecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.6.mlp	Qwen3MLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.6.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.6.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.6.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.layers.6.self_attn	Qwen3Attention	–	–	–
model.layers.6.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.6.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.6.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.6.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.6.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.6.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.7	Qwen3DecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.7.mlp	Qwen3MLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.7.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.7.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.7.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.7.self_attn	Qwen3Attention	–	–	–
model.layers.7.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.7.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.7.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.7.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.7.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.7.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.8	Qwen3DecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.8.mlp	Qwen3MLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.8.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.8.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.8.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.8.self_attn	Qwen3Attention	–	–	–
model.layers.8.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.8.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.8.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.8.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.8.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.8.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.9	Qwen3DecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.9.mlp	Qwen3MLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.9.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.9.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.9.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.9.self_attn	Qwen3Attention	–	–	–
model.layers.9.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.9.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.9.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.9.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.9.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.9.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.10	Qwen3DecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.10.mlp	Qwen3MLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.10.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.10.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.10.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.10.self_attn	Qwen3Attention	–	–	–
model.layers.10.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.10.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.10.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.10.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.10.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.10.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.11	Qwen3DecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.11.mlp	Qwen3MLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.11.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.11.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728

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Layer path	Type	Shape	Dtype	Params
model.layers.11.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.11.self_attn	Qwen3Attention	–	–	–
model.layers.11.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.11.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.11.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.11.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.11.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.11.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.12	Qwen3DecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.12.mlp	Qwen3MLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.12.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.12.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.12.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.12.self_attn	Qwen3Attention	–	–	–
model.layers.12.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.12.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.12.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.12.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.12.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.12.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.13	Qwen3DecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.13.mlp	Qwen3MLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.13.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.13.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.13.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.13.self_attn	Qwen3Attention	–	–	–
model.layers.13.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.13.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.13.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.13.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.13.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.13.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.14	Qwen3DecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.14.mlp	Qwen3MLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.14.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.14.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.14.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.14.self_attn	Qwen3Attention	–	–	–
model.layers.14.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.14.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.14.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.14.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.14.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.14.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.15	Qwen3DecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.15.mlp	Qwen3MLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.15.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.15.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.15.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.15.self_attn	Qwen3Attention	–	–	–
model.layers.15.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.15.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.15.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.15.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.15.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.15.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.16	Qwen3DecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.16.mlp	Qwen3MLP	–	–	–
model.layers.16.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.16.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728

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Layer path	Type	Shape	Dtype	Params
model.layers.16.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.16.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.16.self_attn	Qwen3Attention	–	–	–
model.layers.16.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.16.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.16.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.16.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.16.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.16.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.17	Qwen3DecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.17.mlp	Qwen3MLP	–	–	–
model.layers.17.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.17.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.17.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.17.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.17.self_attn	Qwen3Attention	–	–	–
model.layers.17.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.17.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.17.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.17.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.17.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.17.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.18	Qwen3DecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.18.mlp	Qwen3MLP	–	–	–
model.layers.18.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.18.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.18.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.18.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.18.self_attn	Qwen3Attention	–	–	–
model.layers.18.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.18.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.18.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.18.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.18.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.18.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.19	Qwen3DecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.19.mlp	Qwen3MLP	–	–	–
model.layers.19.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.19.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.19.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.19.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.19.self_attn	Qwen3Attention	–	–	–
model.layers.19.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.19.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.19.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.19.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.19.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.19.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.20	Qwen3DecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.20.mlp	Qwen3MLP	–	–	–
model.layers.20.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.20.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.20.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.20.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.20.self_attn	Qwen3Attention	–	–	–
model.layers.20.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.20.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.20.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.20.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.20.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.20.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.21	Qwen3DecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.21.mlp	Qwen3MLP	–	–	–
model.layers.21.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728

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Layer path	Type	Shape	Dtype	Params
model.layers.21.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.21.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.21.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.21.self_attn	Qwen3Attention	–	–	–
model.layers.21.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.21.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.21.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.21.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.21.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.21.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.22	Qwen3DecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.22.mlp	Qwen3MLP	–	–	–
model.layers.22.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.22.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.22.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.22.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.22.self_attn	Qwen3Attention	–	–	–
model.layers.22.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.22.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.22.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.22.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.22.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.22.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.23	Qwen3DecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.23.mlp	Qwen3MLP	–	–	–
model.layers.23.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.23.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.23.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.23.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.23.self_attn	Qwen3Attention	–	–	–
model.layers.23.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.23.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.23.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.23.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.23.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.23.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.24	Qwen3DecoderLayer	–	–	–
model.layers.24.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.24.mlp	Qwen3MLP	–	–	–
model.layers.24.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.24.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.24.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.24.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.24.self_attn	Qwen3Attention	–	–	–
model.layers.24.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.24.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.24.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.24.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.24.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.24.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.25	Qwen3DecoderLayer	–	–	–
model.layers.25.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.25.mlp	Qwen3MLP	–	–	–
model.layers.25.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.25.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.25.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.25.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.25.self_attn	Qwen3Attention	–	–	–
model.layers.25.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.25.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.25.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.25.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.25.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.25.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.26	Qwen3DecoderLayer	–	–	–
model.layers.26.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.26.mlp	Qwen3MLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.26.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.26.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.26.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.26.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.26.self_attn	Qwen3Attention	–	–	–
model.layers.26.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.26.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.26.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.26.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.26.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.26.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.27	Qwen3DecoderLayer	–	–	–
model.layers.27.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.27.mlp	Qwen3MLP	–	–	–
model.layers.27.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.27.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.27.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.27.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.27.self_attn	Qwen3Attention	–	–	–
model.layers.27.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.27.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.27.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.27.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.27.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.27.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.norm.weight	Qwen3RMSNorm	1024	bfloat16	1,024

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.embed_tokens	155,582,464	155,582,464
model.layers.0.input_layernorm	1,024	155,583,488
model.layers.0.mlp.down_proj	3,145,728	158,729,216
model.layers.0.mlp.gate_proj	3,145,728	161,874,944
model.layers.0.mlp.up_proj	3,145,728	165,020,672
model.layers.0.post_attention_layernorm	1,024	165,021,696
model.layers.0.self_attn.k_norm	128	165,021,824
model.layers.0.self_attn.k_proj	1,048,576	166,070,400
model.layers.0.self_attn.o_proj	2,097,152	168,167,552
model.layers.0.self_attn.q_norm	128	168,167,680
model.layers.0.self_attn.q_proj	2,097,152	170,264,832
model.layers.0.self_attn.v_proj	1,048,576	171,313,408
model.layers.1.input_layernorm	1,024	171,314,432
model.layers.1.mlp.down_proj	3,145,728	174,460,160
model.layers.1.mlp.gate_proj	3,145,728	177,605,888
model.layers.1.mlp.up_proj	3,145,728	180,751,616
model.layers.1.post_attention_layernorm	1,024	180,752,640
model.layers.1.self_attn.k_norm	128	180,752,768
model.layers.1.self_attn.k_proj	1,048,576	181,801,344
model.layers.1.self_attn.o_proj	2,097,152	183,898,496
model.layers.1.self_attn.q_norm	128	183,898,624
model.layers.1.self_attn.q_proj	2,097,152	185,995,776
model.layers.1.self_attn.v_proj	1,048,576	187,044,352
model.layers.2.input_layernorm	1,024	187,045,376
model.layers.2.mlp.down_proj	3,145,728	190,191,104
model.layers.2.mlp.gate_proj	3,145,728	193,336,832
model.layers.2.mlp.up_proj	3,145,728	196,482,560
model.layers.2.post_attention_layernorm	1,024	196,483,584
model.layers.2.self_attn.k_norm	128	196,483,712
model.layers.2.self_attn.k_proj	1,048,576	197,532,288

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Module	Params	Running total
model.layers.2.self_attn.o_proj	2,097,152	199,629,440
model.layers.2.self_attn.q_norm	128	199,629,568
model.layers.2.self_attn.q_proj	2,097,152	201,726,720
model.layers.2.self_attn.v_proj	1,048,576	202,775,296
model.layers.3.input_layernorm	1,024	202,776,320
model.layers.3.mlp.down_proj	3,145,728	205,922,048
model.layers.3.mlp.gate_proj	3,145,728	209,067,776
model.layers.3.mlp.up_proj	3,145,728	212,213,504
model.layers.3.post_attention_layernorm	1,024	212,214,528
model.layers.3.self_attn.k_norm	128	212,214,656
model.layers.3.self_attn.k_proj	1,048,576	213,263,232
model.layers.3.self_attn.o_proj	2,097,152	215,360,384
model.layers.3.self_attn.q_norm	128	215,360,512
model.layers.3.self_attn.q_proj	2,097,152	217,457,664
model.layers.3.self_attn.v_proj	1,048,576	218,506,240
model.layers.4.input_layernorm	1,024	218,507,264
model.layers.4.mlp.down_proj	3,145,728	221,652,992
model.layers.4.mlp.gate_proj	3,145,728	224,798,720
model.layers.4.mlp.up_proj	3,145,728	227,944,448
model.layers.4.post_attention_layernorm	1,024	227,945,472
model.layers.4.self_attn.k_norm	128	227,945,600
model.layers.4.self_attn.k_proj	1,048,576	228,994,176
model.layers.4.self_attn.o_proj	2,097,152	231,091,328
model.layers.4.self_attn.q_norm	128	231,091,456
model.layers.4.self_attn.q_proj	2,097,152	233,188,608
model.layers.4.self_attn.v_proj	1,048,576	234,237,184
model.layers.5.input_layernorm	1,024	234,238,208
model.layers.5.mlp.down_proj	3,145,728	237,383,936
model.layers.5.mlp.gate_proj	3,145,728	240,529,664
model.layers.5.mlp.up_proj	3,145,728	243,675,392
model.layers.5.post_attention_layernorm	1,024	243,676,416
model.layers.5.self_attn.k_norm	128	243,676,544
model.layers.5.self_attn.k_proj	1,048,576	244,725,120
model.layers.5.self_attn.o_proj	2,097,152	246,822,272
model.layers.5.self_attn.q_norm	128	246,822,400
model.layers.5.self_attn.q_proj	2,097,152	248,919,552
model.layers.5.self_attn.v_proj	1,048,576	249,968,128
model.layers.6.input_layernorm	1,024	249,969,152
model.layers.6.mlp.down_proj	3,145,728	253,114,880
model.layers.6.mlp.gate_proj	3,145,728	256,260,608
model.layers.6.mlp.up_proj	3,145,728	259,406,336
model.layers.6.post_attention_layernorm	1,024	259,407,360
model.layers.6.self_attn.k_norm	128	259,407,488
model.layers.6.self_attn.k_proj	1,048,576	260,456,064
model.layers.6.self_attn.o_proj	2,097,152	262,553,216
model.layers.6.self_attn.q_norm	128	262,553,344
model.layers.6.self_attn.q_proj	2,097,152	264,650,496
model.layers.6.self_attn.v_proj	1,048,576	265,699,072
model.layers.7.input_layernorm	1,024	265,700,096
model.layers.7.mlp.down_proj	3,145,728	268,845,824
model.layers.7.mlp.gate_proj	3,145,728	271,991,552
model.layers.7.mlp.up_proj	3,145,728	275,137,280
model.layers.7.post_attention_layernorm	1,024	275,138,304
model.layers.7.self_attn.k_norm	128	275,138,432
model.layers.7.self_attn.k_proj	1,048,576	276,187,008
model.layers.7.self_attn.o_proj	2,097,152	278,284,160
model.layers.7.self_attn.q_norm	128	278,284,288
model.layers.7.self_attn.q_proj	2,097,152	280,381,440

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Module	Params	Running total
model.layers.7.self_attn.v_proj	1,048,576	281,430,016
model.layers.8.input_layernorm	1,024	281,431,040
model.layers.8.mlp.down_proj	3,145,728	284,576,768
model.layers.8.mlp.gate_proj	3,145,728	287,722,496
model.layers.8.mlp.up_proj	3,145,728	290,868,224
model.layers.8.post_attention_layernorm	1,024	290,869,248
model.layers.8.self_attn.k_norm	128	290,869,376
model.layers.8.self_attn.k_proj	1,048,576	291,917,952
model.layers.8.self_attn.o_proj	2,097,152	294,015,104
model.layers.8.self_attn.q_norm	128	294,015,232
model.layers.8.self_attn.q_proj	2,097,152	296,112,384
model.layers.8.self_attn.v_proj	1,048,576	297,160,960
model.layers.9.input_layernorm	1,024	297,161,984
model.layers.9.mlp.down_proj	3,145,728	300,307,712
model.layers.9.mlp.gate_proj	3,145,728	303,453,440
model.layers.9.mlp.up_proj	3,145,728	306,599,168
model.layers.9.post_attention_layernorm	1,024	306,600,192
model.layers.9.self_attn.k_norm	128	306,600,320
model.layers.9.self_attn.k_proj	1,048,576	307,648,896
model.layers.9.self_attn.o_proj	2,097,152	309,746,048
model.layers.9.self_attn.q_norm	128	309,746,176
model.layers.9.self_attn.q_proj	2,097,152	311,843,328
model.layers.9.self_attn.v_proj	1,048,576	312,891,904
model.layers.10.input_layernorm	1,024	312,892,928
model.layers.10.mlp.down_proj	3,145,728	316,038,656
model.layers.10.mlp.gate_proj	3,145,728	319,184,384
model.layers.10.mlp.up_proj	3,145,728	322,330,112
model.layers.10.post_attention_layernorm	1,024	322,331,136
model.layers.10.self_attn.k_norm	128	322,331,264
model.layers.10.self_attn.k_proj	1,048,576	323,379,840
model.layers.10.self_attn.o_proj	2,097,152	325,476,992
model.layers.10.self_attn.q_norm	128	325,477,120
model.layers.10.self_attn.q_proj	2,097,152	327,574,272
model.layers.10.self_attn.v_proj	1,048,576	328,622,848
model.layers.11.input_layernorm	1,024	328,623,872
model.layers.11.mlp.down_proj	3,145,728	331,769,600
model.layers.11.mlp.gate_proj	3,145,728	334,915,328
model.layers.11.mlp.up_proj	3,145,728	338,061,056
model.layers.11.post_attention_layernorm	1,024	338,062,080
model.layers.11.self_attn.k_norm	128	338,062,208
model.layers.11.self_attn.k_proj	1,048,576	339,110,784
model.layers.11.self_attn.o_proj	2,097,152	341,207,936
model.layers.11.self_attn.q_norm	128	341,208,064
model.layers.11.self_attn.q_proj	2,097,152	343,305,216
model.layers.11.self_attn.v_proj	1,048,576	344,353,792
model.layers.12.input_layernorm	1,024	344,354,816
model.layers.12.mlp.down_proj	3,145,728	347,500,544
model.layers.12.mlp.gate_proj	3,145,728	350,646,272
model.layers.12.mlp.up_proj	3,145,728	353,792,000
model.layers.12.post_attention_layernorm	1,024	353,793,024
model.layers.12.self_attn.k_norm	128	353,793,152
model.layers.12.self_attn.k_proj	1,048,576	354,841,728
model.layers.12.self_attn.o_proj	2,097,152	356,938,880
model.layers.12.self_attn.q_norm	128	356,939,008
model.layers.12.self_attn.q_proj	2,097,152	359,036,160
model.layers.12.self_attn.v_proj	1,048,576	360,084,736
model.layers.13.input_layernorm	1,024	360,085,760
model.layers.13.mlp.down_proj	3,145,728	363,231,488

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Module	Params	Running total
model.layers.13.mlp.gate_proj	3,145,728	366,377,216
model.layers.13.mlp.up_proj	3,145,728	369,522,944
model.layers.13.post_attention_layernorm	1,024	369,523,968
model.layers.13.self_attn.k_norm	128	369,524,096
model.layers.13.self_attn.k_proj	1,048,576	370,572,672
model.layers.13.self_attn.o_proj	2,097,152	372,669,824
model.layers.13.self_attn.q_norm	128	372,669,952
model.layers.13.self_attn.q_proj	2,097,152	374,767,104
model.layers.13.self_attn.v_proj	1,048,576	375,815,680
model.layers.14.input_layernorm	1,024	375,816,704
model.layers.14.mlp.down_proj	3,145,728	378,962,432
model.layers.14.mlp.gate_proj	3,145,728	382,108,160
model.layers.14.mlp.up_proj	3,145,728	385,253,888
model.layers.14.post_attention_layernorm	1,024	385,254,912
model.layers.14.self_attn.k_norm	128	385,255,040
model.layers.14.self_attn.k_proj	1,048,576	386,303,616
model.layers.14.self_attn.o_proj	2,097,152	388,400,768
model.layers.14.self_attn.q_norm	128	388,400,896
model.layers.14.self_attn.q_proj	2,097,152	390,498,048
model.layers.14.self_attn.v_proj	1,048,576	391,546,624
model.layers.15.input_layernorm	1,024	391,547,648
model.layers.15.mlp.down_proj	3,145,728	394,693,376
model.layers.15.mlp.gate_proj	3,145,728	397,839,104
model.layers.15.mlp.up_proj	3,145,728	400,984,832
model.layers.15.post_attention_layernorm	1,024	400,985,856
model.layers.15.self_attn.k_norm	128	400,985,984
model.layers.15.self_attn.k_proj	1,048,576	402,034,560
model.layers.15.self_attn.o_proj	2,097,152	404,131,712
model.layers.15.self_attn.q_norm	128	404,131,840
model.layers.15.self_attn.q_proj	2,097,152	406,228,992
model.layers.15.self_attn.v_proj	1,048,576	407,277,568
model.layers.16.input_layernorm	1,024	407,278,592
model.layers.16.mlp.down_proj	3,145,728	410,424,320
model.layers.16.mlp.gate_proj	3,145,728	413,570,048
model.layers.16.mlp.up_proj	3,145,728	416,715,776
model.layers.16.post_attention_layernorm	1,024	416,716,800
model.layers.16.self_attn.k_norm	128	416,716,928
model.layers.16.self_attn.k_proj	1,048,576	417,765,504
model.layers.16.self_attn.o_proj	2,097,152	419,862,656
model.layers.16.self_attn.q_norm	128	419,862,784
model.layers.16.self_attn.q_proj	2,097,152	421,959,936
model.layers.16.self_attn.v_proj	1,048,576	423,008,512
model.layers.17.input_layernorm	1,024	423,009,536
model.layers.17.mlp.down_proj	3,145,728	426,155,264
model.layers.17.mlp.gate_proj	3,145,728	429,300,992
model.layers.17.mlp.up_proj	3,145,728	432,446,720
model.layers.17.post_attention_layernorm	1,024	432,447,744
model.layers.17.self_attn.k_norm	128	432,447,872
model.layers.17.self_attn.k_proj	1,048,576	433,496,448
model.layers.17.self_attn.o_proj	2,097,152	435,593,600
model.layers.17.self_attn.q_norm	128	435,593,728
model.layers.17.self_attn.q_proj	2,097,152	437,690,880
model.layers.17.self_attn.v_proj	1,048,576	438,739,456
model.layers.18.input_layernorm	1,024	438,740,480
model.layers.18.mlp.down_proj	3,145,728	441,886,208
model.layers.18.mlp.gate_proj	3,145,728	445,031,936
model.layers.18.mlp.up_proj	3,145,728	448,177,664
model.layers.18.post_attention_layernorm	1,024	448,178,688

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Module	Params	Running total
model.layers.18.self_attn.k_norm	128	448,178,816
model.layers.18.self_attn.k_proj	1,048,576	449,227,392
model.layers.18.self_attn.o_proj	2,097,152	451,324,544
model.layers.18.self_attn.q_norm	128	451,324,672
model.layers.18.self_attn.q_proj	2,097,152	453,421,824
model.layers.18.self_attn.v_proj	1,048,576	454,470,400
model.layers.19.input_layernorm	1,024	454,471,424
model.layers.19.mlp.down_proj	3,145,728	457,617,152
model.layers.19.mlp.gate_proj	3,145,728	460,762,880
model.layers.19.mlp.up_proj	3,145,728	463,908,608
model.layers.19.post_attention_layernorm	1,024	463,909,632
model.layers.19.self_attn.k_norm	128	463,909,760
model.layers.19.self_attn.k_proj	1,048,576	464,958,336
model.layers.19.self_attn.o_proj	2,097,152	467,055,488
model.layers.19.self_attn.q_norm	128	467,055,616
model.layers.19.self_attn.q_proj	2,097,152	469,152,768
model.layers.19.self_attn.v_proj	1,048,576	470,201,344
model.layers.20.input_layernorm	1,024	470,202,368
model.layers.20.mlp.down_proj	3,145,728	473,348,096
model.layers.20.mlp.gate_proj	3,145,728	476,493,824
model.layers.20.mlp.up_proj	3,145,728	479,639,552
model.layers.20.post_attention_layernorm	1,024	479,640,576
model.layers.20.self_attn.k_norm	128	479,640,704
model.layers.20.self_attn.k_proj	1,048,576	480,689,280
model.layers.20.self_attn.o_proj	2,097,152	482,786,432
model.layers.20.self_attn.q_norm	128	482,786,560
model.layers.20.self_attn.q_proj	2,097,152	484,883,712
model.layers.20.self_attn.v_proj	1,048,576	485,932,288
model.layers.21.input_layernorm	1,024	485,933,312
model.layers.21.mlp.down_proj	3,145,728	489,079,040
model.layers.21.mlp.gate_proj	3,145,728	492,224,768
model.layers.21.mlp.up_proj	3,145,728	495,370,496
model.layers.21.post_attention_layernorm	1,024	495,371,520
model.layers.21.self_attn.k_norm	128	495,371,648
model.layers.21.self_attn.k_proj	1,048,576	496,420,224
model.layers.21.self_attn.o_proj	2,097,152	498,517,376
model.layers.21.self_attn.q_norm	128	498,517,504
model.layers.21.self_attn.q_proj	2,097,152	500,614,656
model.layers.21.self_attn.v_proj	1,048,576	501,663,232
model.layers.22.input_layernorm	1,024	501,664,256
model.layers.22.mlp.down_proj	3,145,728	504,809,984
model.layers.22.mlp.gate_proj	3,145,728	507,955,712
model.layers.22.mlp.up_proj	3,145,728	511,101,440
model.layers.22.post_attention_layernorm	1,024	511,102,464
model.layers.22.self_attn.k_norm	128	511,102,592
model.layers.22.self_attn.k_proj	1,048,576	512,151,168
model.layers.22.self_attn.o_proj	2,097,152	514,248,320
model.layers.22.self_attn.q_norm	128	514,248,448
model.layers.22.self_attn.q_proj	2,097,152	516,345,600
model.layers.22.self_attn.v_proj	1,048,576	517,394,176
model.layers.23.input_layernorm	1,024	517,395,200
model.layers.23.mlp.down_proj	3,145,728	520,540,928
model.layers.23.mlp.gate_proj	3,145,728	523,686,656
model.layers.23.mlp.up_proj	3,145,728	526,832,384
model.layers.23.post_attention_layernorm	1,024	526,833,408
model.layers.23.self_attn.k_norm	128	526,833,536
model.layers.23.self_attn.k_proj	1,048,576	527,882,112
model.layers.23.self_attn.o_proj	2,097,152	529,979,264

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Module	Params	Running total
model.layers.23.self_attn.q_norm	128	529,979,392
model.layers.23.self_attn.q_proj	2,097,152	532,076,544
model.layers.23.self_attn.v_proj	1,048,576	533,125,120
model.layers.24.input_layernorm	1,024	533,126,144
model.layers.24.mlp.down_proj	3,145,728	536,271,872
model.layers.24.mlp.gate_proj	3,145,728	539,417,600
model.layers.24.mlp.up_proj	3,145,728	542,563,328
model.layers.24.post_attention_layernorm	1,024	542,564,352
model.layers.24.self_attn.k_norm	128	542,564,480
model.layers.24.self_attn.k_proj	1,048,576	543,613,056
model.layers.24.self_attn.o_proj	2,097,152	545,710,208
model.layers.24.self_attn.q_norm	128	545,710,336
model.layers.24.self_attn.q_proj	2,097,152	547,807,488
model.layers.24.self_attn.v_proj	1,048,576	548,856,064
model.layers.25.input_layernorm	1,024	548,857,088
model.layers.25.mlp.down_proj	3,145,728	552,002,816
model.layers.25.mlp.gate_proj	3,145,728	555,148,544
model.layers.25.mlp.up_proj	3,145,728	558,294,272
model.layers.25.post_attention_layernorm	1,024	558,295,296
model.layers.25.self_attn.k_norm	128	558,295,424
model.layers.25.self_attn.k_proj	1,048,576	559,344,000
model.layers.25.self_attn.o_proj	2,097,152	561,441,152
model.layers.25.self_attn.q_norm	128	561,441,280
model.layers.25.self_attn.q_proj	2,097,152	563,538,432
model.layers.25.self_attn.v_proj	1,048,576	564,587,008
model.layers.26.input_layernorm	1,024	564,588,032
model.layers.26.mlp.down_proj	3,145,728	567,733,760
model.layers.26.mlp.gate_proj	3,145,728	570,879,488
model.layers.26.mlp.up_proj	3,145,728	574,025,216
model.layers.26.post_attention_layernorm	1,024	574,026,240
model.layers.26.self_attn.k_norm	128	574,026,368
model.layers.26.self_attn.k_proj	1,048,576	575,074,944
model.layers.26.self_attn.o_proj	2,097,152	577,172,096
model.layers.26.self_attn.q_norm	128	577,172,224
model.layers.26.self_attn.q_proj	2,097,152	579,269,376
model.layers.26.self_attn.v_proj	1,048,576	580,317,952
model.layers.27.input_layernorm	1,024	580,318,976
model.layers.27.mlp.down_proj	3,145,728	583,464,704
model.layers.27.mlp.gate_proj	3,145,728	586,610,432
model.layers.27.mlp.up_proj	3,145,728	589,756,160
model.layers.27.post_attention_layernorm	1,024	589,757,184
model.layers.27.self_attn.k_norm	128	589,757,312
model.layers.27.self_attn.k_proj	1,048,576	590,805,888
model.layers.27.self_attn.o_proj	2,097,152	592,903,040
model.layers.27.self_attn.q_norm	128	592,903,168
model.layers.27.self_attn.q_proj	2,097,152	595,000,320
model.layers.27.self_attn.v_proj	1,048,576	596,048,896
model.norm	1,024	596,049,920